CSE382 — Hotel Booking Demand ML Project

Exploratory Data Analysis (EDA) Progress Report

Dataset: H2.csv

1. Introduction

The exploratory data analysis (EDA) stage was conducted to understand the Hotel Booking Demand dataset, assess its data quality, explore relationships between features, and identify potential issues before developing the required machine-learning models.

The project includes three machine-learning tasks:

- **Classification:** Predict whether a booking will be canceled using `IsCanceled`.
- **Regression:** Predict the Average Daily Rate using ADR.
- **Clustering:** Identify groups of similar bookings without a predefined target variable.

The original dataset has been preserved without modifications. Data cleaning and preparation will be performed separately in a dedicated notebook.

2. Dataset Overview

Property	Finding
Dataset	`H2.csv`
Total records	79,330
Total features	31
Arrival years	2015–2017
Memory usage	Approximately 28.6 MB
Classification target	`IsCanceled`
Regression target	ADR
Clustering target	None
Original dataset modified	No

The dataset contains booking details, arrival information, customer characteristics, reservation attributes, and cancellation outcomes. These features provide a foundation for investigating booking behavior and developing predictive models.

3. Data Quality Assessment

3.1 Missing Values

The initial inspection identified missing values in two columns.

Column	Missing Values
`Children`	4
`Country`	24

Although the number of missing values is small relative to the dataset size, these values will need to be handled during data preparation. The treatment will depend on the meaning of each feature and the requirements of the relevant model.

3.2 Duplicate Records

The dataset contains a substantial number of exact duplicate rows.

Metric	Result
Original records	79,330
Duplicate occurrences	25,902
Percentage of original records	32.65%
Records after removing duplicates	53,428

A duplicate occurrence represents a row that matches an earlier row across the inspected columns. However, identical rows do not necessarily represent the same real-world booking.

The inspected dataset does not provide a unique booking ID that would allow us to confirm whether identical rows belong to the same booking or to different legitimate bookings with identical recorded attributes.

> **Decision: No final deduplication policy has been selected.** We will preserve the original data and compare the results obtained using both versions:
> 1. The original dataset, retaining duplicate rows.
> 2. A deduplicated copy, with exact duplicate rows removed.

We will evaluate how this choice affects model performance and the reliability of the results before deciding which approach is more appropriate. The deduplicated dataset will be treated as an experimental alternative, not automatically as the correct dataset.

3.3 Comparison Before and After Duplicate Removal

A separate copy of the original dataset was created with exact duplicates removed to measure the effect of deduplication on the dataset and its target distribution.

Metric	Before Removal	After Removal
Total records	79,330	53,428
Duplicate occurrences remaining	25,902	0
Not canceled bookings	46,228 (58.27%)	37,385 (69.96%)
Canceled bookings	33,102 (41.73%)	16,043 (30.04%)
Missing `Children` values	4	4
Missing `Country` values	24	6

Key findings

- Removing duplicates reduced the dataset by 25,902 records, approximately 32.65%.
- The cancellation rate decreased from 41.73% to 30.04%.
- The proportion of non-canceled bookings increased from 58.27% to 69.96%.
- Missing `Country` values decreased from 24 to 6, while missing `Children` values remained unchanged.

These changes show that deduplication has a substantial effect on both dataset size and class distribution.

Planned experiment: Train and evaluate comparable models using both dataset versions, applying the same appropriate evaluation methodology. Compare the results to understand how retaining or removing duplicates affects predictive performance and generalization. This experiment will inform our decision; a performance difference alone will not prove that one version is inherently more correct.

4. Classification Analysis — `IsCanceled`

The classification task aims to predict whether a booking will be canceled using relevant booking information.

4.1 Target Distribution

The distribution of `IsCanceled` was examined to understand the balance between the two classes.

Booking Outcome	Count	Percentage
Not canceled (0)	46,228	58.27%
Canceled (1)	33,102	41.73%
Total	**79,330**	**100%**

Both classes have substantial representation in the original dataset. However, the class distribution changes after duplicate removal, so the dataset version used for training and evaluation must be documented.

4.2 Cancellation Rate by Lead Time

Bookings were grouped by lead time to examine how the time between booking and arrival relates to cancellation behavior.

Lead Time	Cancellation Rate	Number of Records
0–7 days	12.20%	10,808
8–30 days	30.91%	12,554
31–90 days	39.91%	20,797
91–180 days	47.99%	18,223
181–365 days	62.72%	14,244
366+ days	71.12%	2,704

The cancellation rate generally increases as lead time becomes longer. This makes `LeadTime` a potentially useful predictor for the classification model.

4.3 Cancellation Rate by Deposit Type

Deposit Type	Cancellation Rate	Number of Records
Non Refund	99.81%	12,868
Refundable	70.00%	20
No Deposit	30.47%	66,442

The Non Refund category has an unusually high observed cancellation rate and warrants further investigation. The Refundable category contains only 20 records, so its cancellation rate should be interpreted cautiously.

4.4 Cancellation Rate by Market Segment

Market Segment	Cancellation Rate	Number of Records
Undefined	100.00%	2
Groups	68.86%	13,975
Offline TA/TO	42.83%	16,747
Online TA	37.40%	38,748
Aviation	21.94%	237
Corporate	21.47%	2,986
Direct	17.33%	6,093
Complementary	11.81%	542

Cancellation rates vary across market segments. The Undefined category has only two records, so its 100% rate is not a reliable basis for broad conclusions.

4.5 Cancellation Rate by Customer Type

Customer Type	Cancellation Rate	Number of Records
Contract	48.04%	2,300
Transient	45.62%	59,404
Transient-Party	28.10%	17,333
Group	9.90%	293

The observed cancellation rate differs across customer types, suggesting that `CustomerType` may provide useful information for predicting cancellations.

> These relationships are observational and do not establish that any individual feature causes a booking to be canceled. The rates above were calculated using the original dataset, including duplicates.

5. Cross-Tabulation and Data Leakage

A cross-tabulation was performed between `ReservationStatus` and `IsCanceled` to examine whether the reservation status reveals the classification target.

Reservation Status	Not Canceled (0)	Canceled (1)	Total
Canceled	0	32,186	32,186
Check-Out	46,228	0	46,228
No-Show	0	916	916
Total	**46,228**	**33,102**	**79,330**

The table shows that `ReservationStatus` directly reveals the value of `IsCanceled` in this dataset. Including it as a predictor would cause data leakage and produce misleading evaluation results.

Feature	Treatment for Classification
`IsCanceled`	Use as the target variable
`ReservationStatus`	Exclude from input features
`ReservationStatusDate`	Exclude from input features

Other features will be assessed according to whether they would realistically be available at the time a cancellation prediction is made.

6. Regression Analysis — ADR

The regression task aims to predict **ADR (Average Daily Rate)**, as specified in the project assignment. Initial analysis was conducted to understand its distribution and identify potentially unusual observations.

6.1 ADR Summary Statistics

Statistic	ADR Value
Mean	105.30
Minimum	0
25th percentile	79.20
Median	99.90
75th percentile	126.00
Maximum	5,400.00

The mean is higher than the median, and the maximum is substantially greater than the upper quartile. This suggests a right-skewed distribution with some unusually high values.

The highest ADR value, 5,400, was found in a record associated with a canceled booking. This observation requires further investigation because ADR for canceled bookings may not represent a price that was actually paid.

Considerations for regression:

- Investigate extreme ADR values before deciding how to treat them.
- Examine whether ADR values are meaningful for canceled bookings.
- Ensure ADR is used as the target and is not included among the input features.
- Evaluate models using MAE, RMSE, and R^2, alongside any other metrics required by the assignment.

ADR will be the regression target. Model development and tuning will take place after data preparation.

The regression task will also be included in the planned comparison between the original and deduplicated datasets.

7. Categorical Feature Overview

The categorical features were inspected to understand their variety and potential preparation requirements.

Feature	Number of Unique Values
`ArrivalDateMonth`	12
`Meal`	4
`Country`	166
`MarketSegment`	8
`DistributionChannel`	5
`ReservedRoomType`	8
`AssignedRoomType`	9
`DepositType`	3
`Agent`	224
`Company`	208
`CustomerType`	4
`ReservationStatus`	3
`ReservationStatusDate`	864

Some categorical features have many distinct values. Features such as `Country`, `Agent`, and `Company` may require particular attention during encoding and feature selection.

The number of unique values does not indicate the number of missing values. Feature usefulness and availability at prediction time will also be considered during preparation.

8. Clustering Considerations

The clustering task will identify groups of bookings with similar characteristics without relying on a predefined target variable.

The dataset contains a mixture of numerical and categorical features, so preparation will be important before clustering. Numerical features may need scaling, while categorical features will require an appropriate representation.

The planned experiment will also compare clustering results using the original and deduplicated datasets. This will help us assess whether repeated rows influence the groups identified by the clustering algorithms.

9. Data Preparation Considerations

The following considerations were identified during EDA:

- **Missing values:** Select appropriate treatments for `Children` and `Country`.
- **Duplicates:** Compare the original and deduplicated datasets before deciding on a policy.
- **Unusual numerical values:** Investigate extreme observations, particularly for ADR.
- **Categorical encoding:** Select suitable methods for categorical features with different numbers of unique values.

- **Feature selection:** Exclude variables that reveal the target or would not be available at prediction time.
- **Model-specific preparation:** Prepare appropriate input features for classification, regression, and clustering.
- **Evaluation reliability:** Use appropriate metrics and avoid data leakage during training and validation.
- **Reproducibility:** Apply comparable preprocessing and evaluation procedures when comparing dataset versions.

All transformations will be carried out in a separate preparation notebook, leaving the original `H2.csv` intact.

10. Conclusion

The exploratory data analysis established an understanding of the dataset's structure, quality, and important relationships. The analysis identified missing values, a substantial number of duplicate records, unusual ADR values, and a direct relationship between `ReservationStatus` and the cancellation target.

The observed cancellation patterns suggest that features such as `LeadTime`, `DepositType`, and `CustomerType` may be useful for classification. ADR has been confirmed as the regression target, while clustering will focus on discovering groups of similar bookings.

A key unresolved issue is the treatment of duplicate records. Because the dataset does not provide a unique booking ID in the inspected columns, identical rows may represent either repeated records or separate legitimate bookings. We therefore cannot justify removing them automatically.

The next stage will include experiments using both the original and deduplicated dataset versions. Their effects on classification, regression, and clustering results will be evaluated before a final decision is made. Throughout this process, the original `H2.csv` will remain unchanged.